Chapter 1

Demand Estimation and Forecasting

1.1 Introduction

Demand forecasting is an important tool to make decisions such as financial projections, capacity planning and allocation of resources for a productive activity. The demand for any product, service or good in the future is a random variable. That is, it is unknowable before it is realized. The demand that already occurred is called *realized demand* or can also be called historical demand.

In fact, only the past is really known, everything happening in the future has to be guessed. In these lectures notes we deal with the most common practices in *guessing* the future quantities of the demand. Mathematically, it is useful to model the demand as a random variable. Random variables are a mathematical construct to capture events and their randomness. We will cover the basics building up the necessary knowledge to manipulate and eventually model and estimate demand in the future.

1.2 The Demand as a Random Variable

Suppose we want to model the demand quantity once in the future (say, next week or next month or next year). We denote this random quantity by the capital letter D. This random variable D is in fact a function that takes as input a random mechanism ω and outputs an event, in our case a number for the demand[1]. For example, the demand can be a number like $D = 14$ with probability p_{14}, or it could be $D = 5,000$ with a probability $p_{5,000}$.

As normally the demand is a *count* (of products, goods or services) we will assume the demand cannot take negative values. Then, we can describe the probability of any outcome

[1] The space of random mechanism is normally called sample space, and is denoted as the set Ω. Then, formally the random variable is a function mapping $D : \Omega \to E$ to a set of possible events E.

of the demand by a *probability mass function* or pmf as $p_i = \mathsf{P}(D = i)$ for all possible values $i = 0, 1, \ldots$ For the pmf to be correctly defined two things must happen: First, each outcome must occur with a probability between 0 and 1, that is $0 \leq p_i \leq 1$ and adding over all outcomes should add up to 1, that is, $\sum_{i=0}^{\infty} p_i = 1$.

With this framework, we can already solve interesting questions: For example, if D is the quantity of semiconductor chips sold in the next quarter, we can ask *what is the probability that the demand is at least 10 million chips?*. To answer this, the event $\{D > 10^7\}$ is the same as the events $\{D = 10^7 + 1, D = 10^7 + 2, ...\}$, then we can simply do:

$$\mathsf{P}(D > 10^7) = \mathsf{P}(D = 10^7 + 1, D = 10^7 + 2, \ldots) = \sum_{i=10^7+1}^{\infty} p_i, \tag{1.1}$$

Throughout the topic it is often useful to get used to manipulate event sets. For example, in the previous example we exploited the fact that the event $\{D > 10^7\}$ can be written as the union that the demand takes exactly each value greater than 10^7, noting that each of these events is disjoint (it can be a value or another but not two at the same time). We could have also taken an alternative approach by calculating the probability of the *complement* event, that is, that the demand is less or equal to 10^7 and then, given that the probability of all outcomes is 1, subtract $P(D \leq 10^7)$. In other words:

$$\mathsf{P}(D > 10^7) = 1 - \mathsf{P}(D \leq 10^7). \tag{1.2}$$

For a given number i, the probability $F_D(i) = \mathsf{P}(D \leq i)$ is known as the *cumulative mass function* or cmf. The usefulness of the cmf, is related again to event calculus, for example asking the probability that the demand is between 1 and 10 million units. That is, $\mathsf{P}(10^6 < D \leq 10^7)$ can be expressed in terms of the cmf as $F_D(10^7) - F_D(10^6)$.

Another very important quantity is known as the mean, or average. This comes from averaging all possible outcomes for the demand weighted by their probabilities. This quantity is denoted as $\mu_D = \mathsf{E}(D) = \sum_{i=0}^{\infty} ip_i = 0p_0 + 1p_1 + 2p_2 + \ldots$[2] Averages are important for demand estimation, because they provide a scenario that is centered around the more likely outcomes for the demand. For example, when making a resource allocation decision, such as building a factory or a data-center, the capacity should comfortably handle the average demand for the facility. But then, the next immediate question is how much slack capacity should the factory have?

The variance of a random variable is a measure of how much the random variable *deviates* from its mean μ_D. The variance is defined as $\sigma_D^2 = \mathsf{Var}(D) = \mathsf{E}[(D - \mu_D)^2]$, the E operator (or function) denotes the operation of multiplying each possibly outcome by its probability. So, $\mathsf{E}[(D - \mu_D)^2]$ can be computed as $\sum_{i=0}^{\infty}(i - \mu_D)^2 p_i$. The square root of the variance is known as the *standard deviation* as is denoted by the symbol σ_D.

[2] Related to event calculus, the mean μ_D is also equal to $\sum_{i=0}^{\infty} \bar{F}_D(i)$, where $\bar{F}_D(i) = 1 - F_D(i)$. Why is this the case?

1.2.1 Useful Probability Calculations

So far, we have reviewed basic concepts in probability that can answer some questions about the random outcome of the demand. We can go one step further and with some knowledge of the distribution, answer other questions: For example, say that you only know the mean μ_D of the demand, and they ask you, if the average demand is 1 million units, what's the probability that the demand is less than 5 million?

A useful relation known as *Markov's Inequality* states that $\mathsf{P}(D \geq a) \leq \mu_D/a$, replacing $a = 5 \times 10^6$ tells us that the probability that the demand exceeds 5 million units is at most $\mu_D/(5 \times 10^6)$. If the average demand μ_D is 1 million, the probability that it is at least 5 million is at most $\mu_D/a = 10^6/(5 \times 10^6) = 1/5 = 20\%$. Therefore, the probability that the demand is less than 5 million is at least 80% (could be more). That being said, this estimate is rather loose as the inequality is *at most* the value. In the previous example, it could be the case that the probability that the production exceeds 5 million is exactly 0.

Another useful inequality is known as *Chebyschev's Inequality* stating that $\mathsf{P}(|\mu_D - D| \geq k\sigma_D) \leq 1/k^2$. Going back to our previous example, suppose that additionally to knowing that the average demand μ_D is 1 million, we also know that its standard deviation σ_D is 100,000 units. Applying this inequality, the probability that the demand is below 800,000 or above 1,200,000 is at most 25% (using the inequality with $k = 2$). Going back to the previous question about the probability that the demand exceeds 5 million is at most $1/40^2 = 0.06\%$. Even more can be said, for example the probability that the demand exceeds 2 million is less than $1/100 = 1\%$. The moral of this, is that with just the mean μ_D and the standard deviation of the demand σ_D there is already a lot of useful information that can be inferred about the demand without knowing the full probability distribution of all its outcomes.

1.2.2 Side Information

The demand is clearly influenced by other variables (random or deterministic): the economic cycle, interest rates, competitors prices, the demand of other products, weather and so on. Having a framework to understand how a variable affects the demand is crucial to better estimate and anticipate the demand.

Suppose you are in charge of the production of the Airpods wireless earphones. Intuitively, the demand of Airpods is *positively* related to the demand of other products in the iOS ecosystem (such as laptops, iPhones, etc) and *negatively* to competitor products, such as Android headphones. Moreover, the demand for Airpods should not be affected by the demand of apples or oranges.

Ideally, it would be great to have a mathematical function g, that given an outcome of a related random variable $X = x$ gives us the best estimate of the random demand D. You

can imagine X as the demand for iPhones in the AirPods example (as a significant portion of AirPods users own iPhones). Then, we would like to find the function g that in average gives the closest prediction of the demand D. One such formulation is

$$\min_g \mathsf{E}\left[(g(X) - D)^2\right], \tag{1.3}$$

This way of thinking is a strategy that we will repeatedly use in the exposition of the topic. That is, to ask what's the best possible data-driven way to make guesses about the random nature of the demand. Assuming X can take $x_1, x_2, \ldots$ different values. The expected value $\mathsf{E}\left[(g(X) - D)^2\right]$, taking $p_{ix} = \mathsf{P}(D = i, X = x)$ can be written as:

$$\mathsf{E}\left[(g(X) - D)^2\right] = \sum_{i=0}^{\infty} \sum_{x=x_1,\ldots} (g(x) - i)^2 p_{ix}, \tag{1.4}$$

The strategy to solve a seemingly complex problem like this is to break down a complex problem into smaller and easier problems to solve (a well known paradigm in Computer Science known as divide-and-conquer). Then, the first observation is that the solution only depends on the values of the random variable X. That is, for a given $X = x$, the function $g(x)$ is a single value guessing the most likely outcome of the demand D. Taking only a single value of $X = x$ gives the expression $\sum_{i=0}^{\infty}(g(x) - i)^2 p_{ix}$. This expression has a single unknown $g(x)$, which is a convex function[3]. Taking derivatives and equating to 0 leads to the following expression for $g(x)$:

$$g(x) = \sum_{i=0}^{\infty} i \frac{p_{ix}}{p_x} = \mathsf{E}(D|X = x),$$

This expression intuitively says, that the best predictor for the demand D when the side variable X is equal to x is the expected value of the demand given that $X = x$. This is the so-called *conditional expectation*, which can be seen as simply taking the well known expected value with new probabilities p_{ix}/p_x representing the likelihood of each outcome of the demand given a certain outcome for the side variable X. The denominator p_x scales the new probabilities p_{ix} of the demand so that they add up to 1, because instead of considering all outcomes of X, only the case where $X = x$ is considered.

Example 1.1 (Hydroelectric Plant). Suppose you are managing a hydroelectric plant supplying electric energy to a major city. X represents whether or not it is an extremely hot day ($X = 1$ means it is extremely hot, and $X = 0$ means it is not), extremely hot days occur rarely, say $\mathsf{P}(X = 1) = p_h = 5\%$. But when they occur, the demand for electricity is very high (due to the usage of Air Conditioner and other appliances). The conditional distribution of $(D|X)$ is given by $\mathsf{P}(D = 5\text{Mw}|X = 0) = \mathsf{P}(D = 15\text{Mw}|X = 1) = 1$. Given this distribution, what is

[3] A sum of convex functions is convex.

the average demand μ_D of the power plant? The average demand of the power plant is:

$$\mu_D = \mathsf{E}[D] = \mathsf{E}[\mathsf{E}[D|X]] = 5\text{Mw}(1 - p_h) + 15\text{Mw}p_h = 5.5\text{Mw}. \tag{1.5}$$

This relationship clearly shows one of the main ideas of this course: While the demand in average is relatively stable at a level of 5.5Mw most days, with a small probability it can jump to almost a 300% increase if it's a very hot day. If the power plant does not have the capacity to service the demand, it can be a catastrophic event for the city, since many crucial services including hospitals and other vital supply chains rely on electricity. Hedging against rare events is the whole point of a resilient supply chain.

1.3 Demand Evolution over Time

So far, we have presented a static view of the demand as random variable. While useful as a first order tool of analysis, it is easy to imagine that the demand of many products is in fact time-dependent: Ice cream sales peak in the summer, airline traffic peaks around holidays and restaurants demand has modes around lunch and dinner. Similar to last example of the previous section, averaging without side information can generate misleading estimates for the demand. Therefore, if the demand is time-dependent it makes sense to include time in its modeling.

Time is represented as a variable $t = 0, 1, 2, \ldots$ where each value of t represents a moment in time. The unit of time can be taken as seconds, hours, days or years. Depending on the context of the specific supply chain decision: For example, in analysing the demand behaviour in an online store, it makes sense to have relatively small units of time. Conversely, when planning capacities or resources for facilities, it makes sense to consider the demand over years. Later it will also become evident that there is a trade-off between the historical amount of data available and the quality of the estimation of the demand.

Incorporating time into the demand estimation leads to indexing the demand random variable by time as D_t, denoting the random demand at time t. In most scenarios, we have access to the demand up to some point in time in the past. For example, suppose that you are given the monthly historical demand for the last 5 years. Chronologically, these are 60 datapoints $d_0, d_1, d_2, \cdots, d_{59}$. Data like this can be denoted compactly as $\{d_s\}_{s=0}^{t}$ (imagine a spreadsheet with a single column and a row for each entry of the demand at every date). The distinction between the capital and lowercase symbols for the demand denote the random variable D_t and certain outcome for it d_t. This is important, as even though the past is a realization of the demand, the outcome was a result of the probabilistic nature of the demand (for example, a month of low demand could be caused by an unexpected shock, and it might provide information for future outcomes of the demand).

Random variables that evolve over time are referred as *Stochastic Processes*. We present

a couple of examples to get a feel for stochastic processes. One of the simplest models is the following:

$$D_{t+1} = \alpha + \beta D_t + V_t, \tag{1.6}$$

We call this the *Base Model* as it will allow us to model some common behaviours of the demand over time. In this model $\alpha, \beta \in \mathbb{R}$ are two constants while V_t is an independent random variable. In plain words, the equation says that the demand at time $t+1$ is equal to a constant α plus a term multiplying the previous period demand D_t, plus a random quantity V_t (normally this quantity expresses what's commonly referred as an error or simply unexplained variation of the demand). This model is related to the previous one (independent of time) when $\beta = 0$[4].

To see the dependence over time, assume that at time 0, there is an initial demand $D_0 \geq 0$. $D_1 = \alpha+\beta D_0+V_0$, and $D_2 = \alpha+\beta D_1+V_1 = \alpha+\beta(\alpha+\beta D_0+V_0)+V_1 = \alpha+\beta\alpha+V_1+\alpha V_0+\beta^2 D_0$. Continuing this iterative process leads to the following expression:

$$D_{t+1} = \sum_{j=0}^{t-1}(\alpha + V_{t-j})\beta^j + \beta^t D_0, \tag{1.7}$$

Assuming V_t for $t = 0, 1, \ldots$ has identical and independent distribution with $\mathsf{E}(V_t) = 0$. Taking expectations yields the following:

$$\mathsf{E}(D_{t+1}) = \alpha\frac{1-\beta^t}{1-\beta} + \beta^t\mathsf{E}(D_0), \tag{1.8}$$

The first term of the right hand side comes from the finite sum of a geometric series[5]. Then, the expected value of the demand depends on the expected value of the initial demand, plus a term that converges to $\alpha/(1-\beta)$ at t goes to ∞. Moreover, this convergence only occurs if $|\beta| < 1$. If $|\beta| > 1$ the demand explodes as time goes by (think why this is the case). When $\beta = 1$ with $\mathsf{P}(V_t = 1) = \mathsf{P}(V_t = -1) = 1/2$. The model retrieved is called a *Random Walk*. Among other interesting properties, random walks drift away indefinitely to $\pm\infty$ and can also be negative. With a small modification, of taking $\ln(D_t)$ instead of D_t a *Geometric Random Walk* can be used to model the demand.

1.3.1 Modeling Seasonality and Time-varying Factors

Another common feature of the demand is *seasonality*. Seasonality is defined as the fluctuation of the demand caused by the passage of time or certain time-dependent events (such as holidays or weather variations). Think of holidays and demand that changes due to weather or temperature. The most common way to model this is to follow the method in the weather example of the previous section, by adding an extra variable codifying the side-information of interest. In the

[4] If V_t is a random variable so is $\alpha + V_t$. In fact, constants are random variables with their probability mass concentrated at a single value.

[5] Take $s_t = 1 + \beta + \cdots + \beta^t$ and $\beta s_t = \beta + \cdots + \beta^{t+1}$, subtracting the two equations yields the result shown.

case of seasonality, this can be achieved by adding binary variables that are equal to 1 when t is equal to that time. For example, think of daily data where we can to capture day of week effect on the demand (think of a restaurant or retail business). A way to achieve this, is to modify the α term in Equation (1.6) as:

$$\alpha_t = \sum_{d=0}^{s-1} \mathbf{1}\{t \bmod s = d\}\gamma_d, \tag{1.9}$$

In this case, the indicator $\mathbf{1}\{t \bmod s = d\}$ is only one every s observations. In the days of the week example, $s = 7$ as there are 7 days in a week. Putting this expression back in the base model yields:

$$D_{t+1} = \sum_{d=0}^{s-1} \mathbf{1}\{t \bmod s = d\}\gamma_d + \beta D_t + V_t, \tag{1.10}$$

The model is quite similar to the original one (that had only one constant term α while this one has a constant term for every day of the week). In general, we can also include time-varying factors, such as seasons (also encoded in indicators or numerical variables. A clean way to include these factors is to flatten these variables into a vector h_t. For example, letting $h_t = (1, t)'$ retrieves a model with a constant plus a linear trend allows to express the original α_t term as $\alpha_t = Ah_t$ where A in this case is a vector of parameters with the same dimension as the number of time-varying variables, that is $A \in \mathbb{R}^{|h_t|}$. This allows to re-write the base model as:

$$D_{t+1} = \alpha_t + \beta D_t + V_t = Ah_t + \beta D_t + V_t, \tag{1.11}$$

This instance of the base model allows for arbitrary specifications of time-dependent variables.

1.3.2 Other Modeling Specifications

It is easy to imagine that a similar strategy can be employed to add arbitrary random variables or other stochastic processes X_t of variables that are related to the demand D_t. In general, one can imagine that any arbitrary real-valued function can be calibrated to solve the following problem:

$$\min_g \mathsf{E}[(g(D_t, D_{t-1} \dots, X_t, X_{t-1}, \dots) - D_{t+j})^2]. \tag{1.12}$$

It should be evident that this is exactly the same problem in Section 1.2.2 where the optimal solution is $g(D_t, D_{t-1} \dots, X_t, X_{t-1}, \dots) = \mathsf{E}(D_{t+j}|D_t, D_{t-1} \dots, X_t, X_{t-1}, \dots)$. Then, speaking strictly from an optimization point of view, what matters is that the family of functions g can express the conditional expectation of the demand D_{t+j} as a function of the other variables. Whether the model is a simple linear regression or a sophisticated neural network, what matters is that it can capture the shape of the conditional expectation. As we will see next, there is another tension related to the fact that in general the probabilistic distribution of the demand is unknown and only a finite sample of the variables has to be used for estimation.

1.4 Demand Estimation in Practice

So far, we have built a methodological theory to study the evolution of the demand over time. The core message is that the best possible guess for the demand is the conditional expectation of the demand taking into account information provided by other relevant variables. In practice, the situation is significantly different: At best there are limited observations of the historical demand (or no information at all in the worst case). The side-variables that affect the demand are unknown or unobservable and in many cases they are endogenous (competitors set prices reactively to their own demand and compete against each other for market domination). The latter situation is more complex and will be studied in a later chapter. In this section, we focus on the case where we have limited data and want to predict the demand in the *short-term*. Here, short-term is used loosely as the foreseeable future.

The main estimation problem for the demand is that its true probabilistic distribution is unknown. Then, the goal is to learn the demand from *sample* observations. Suppose, a historical yearly demand of salmon in a country. This demand should be stable and more or less proportional to the population of the country, to account for the population fluctuation, consider the *per capita* demand. This historical demand is represented by the observations $\{d_s\}_{s=1}^t$. Estimating the average demand from samples is calculated as:

$$\mu_D(t) = \frac{1}{t}\sum_{s=1}^{t} d_s, \tag{1.13}$$

This average converges to a stable (constant value) in various cases: For example, when the demand samples are *independent and identically distributed*, or i.i.d. Meaning that they come from the same probabilistic distribution, and the previous values do not affect the probability of new outcomes of the demand. One can see getting observations of the demand as sampling the outcome of an experiment repeatedly and measuring the demand after each trial.

It is important to determine how reliable this estimate is, especially when managing a supply chain as making resource allocation decisions in the future can be extremely costly. Probability theory has two answers regarding how trusted this estimator can be: The *Law of Large Numbers* and the *Central Limit Theorem*.

The Law of Large Numbers (LLN) states that for a large enough t, the average demand (sampled i.i.d. with finite mean and variance) $\mu_D(t)$ converges to the true average value μ_D. That is, $\mu_D(t) \to \mu_D$ as $t \to \infty$. As mentioned before, this is important because there is a probabilistic guarantee that sampling the demand i.i.d. and averaging its values over time recovers the true average demand.

The Central Limit Theorem (CLT) goes a step further and provides a probability distribution for the randomness of the demand. Given an i.i.d. sample of the demand $\{d_s\}_{s=1}^t$. The random variable defined by $\frac{\sqrt{t}(\mu_D(t)-\mu_D)}{\sigma_D}$ converges in distribution to a random variable Z distributed

as a standard normal random variable[6].

Example 1.2 (Rice Storage Levels). The government of Japan asks you to conduct a study to determine what level of rice reserves they should hold. Their first question is how many years t of data they should look at to get a 95% confidence estimate of the average demand μ_D that is within 10% of the true value, using the sample estimate $\mu_D(t)$. Suppose you are told that the standard deviation of the demand σ_D is 30% of the mean μ_D. That is, $\sigma_D = \mu_D 30\%$.

Using the CLT we get that the sample average demand is equal in distribution to:

$$\mu_D(t) \stackrel{\text{d}}{=} \frac{\sigma_D}{\sqrt{t}} Z + \mu_D, \tag{1.14}$$

Dividing by μ_D yields the percentage error $\frac{\sigma_D}{\mu_D\sqrt{t}} Z + 1$. Where $\frac{\sigma_D}{\mu_D\sqrt{t}} Z$ is the error in terms of the demand, which can be positive or negative. Then, for this expression to within 10% of the true value with 95% confidence we need the following to be true:

$$\mathsf{P}\left(\left|\frac{\sigma_D}{\mu_D\sqrt{t}} Z\right| \leq 10\%\right) = 95\%, \tag{1.15}$$

Using the fact that $\sigma_D/\mu_D = 30\%$, the probability can be rearranged to $\mathsf{P}(|Z| \leq \sqrt{t}/3) = 1 - \mathsf{P}(|Z| > \sqrt{t}/3) = 95\%$ or $\mathsf{P}(Z > \sqrt{t}/3) = 5\%/2 = 2.5\%$. That is, the tail cdf of the standard normal distribution $\bar{\Phi}(\sqrt{t}/3) = 2.5\%$. Applying the inverse[7] tail cdf yields:

$$\sqrt{t}/3 = \bar{\Phi}^{-1}(2.5\%) = 1.96. \tag{1.16}$$

With this, we get our answer that the number of observations t needed for the sample average to be within 10% of the mean is at least $t = (3(1.96))^2 = 34.5$ years.

In general, the above example is the best possible case. In probability textbooks, a commonly encountered rule is that the sample size for estimation should be at least 30 observations. As both the LLN and the CLT relied on the i.i.d. property of the samples for convergence.

To illustrate this, consider a random demand that follows the following demand process $D_{t+1} = D_t + c + \sigma Z$, with $D_0 = 0$. Note that this process is an instance of the base model with $\alpha = c$, $\beta = 1$ and $V_t = \sigma Z$. In this case, $c + \sigma Z$ is called a normal random variable with mean c and variance σ^2 (likewise, a normal random variable can be transformed in a standard normal by subtracting its mean and dividing by σ. This operation is useful to make computations always in terms of standard normal random variables).

[6] A standard normal random variable has probability density function (pdf) $f(x) = e^{-x^2/2}/\sqrt{2\pi}$. A pdf is the continuous analogous to the pmf in the sense that for a random variable X with density $f(x)$ the probability $\mathsf{P}[X \in (x, x + \delta_x)] = f(x)\delta_x$ for a small $\delta_x > 0$.

[7] An inverse function $f^{-1}(x)$ is a function such that $f^{-1}(f(x)) = x$. For example, the inverse function of $f(x) = e^x$ is $f^{-1}(x) = \ln(x)$.

The demand process is equal to $D_t = ct + \sigma \sum_{i=1}^{t} Z_t$. The expected value of the demand $\mathsf{E}(D_t) = tc$, since each independent standard normal Z_t has mean 0. Likewise, the sample mean (extracted from historical observations of the demand) will be approximately this expected value, that is, $\mu_D(t) \approx tc$. As t increases indefinitely, so will the sample mean, that is, $\mu_D(t) \to \infty$ as $t \to \infty$. So, how can we estimate the demand in these cases?

Most textbooks in Time Series spend a significant amount of their exposition in studying *stationary* stochastic processes. That is, processes that converge to a stable mean in the long-term and do not explode indefinitely. The textbook solution is to apply a transformation to the original series, such that the transformation is stationary. In the previous example, a solution is to consider a new series taking the differences of the demand. For example, consider the difference process $\Delta_t = D_t - D_{t-1} = c + \sigma Z$, which as discussed previously is a normal random variable with mean c and variance σ^2. The sample mean of historical observations of Δ_t should be the same as sampling standard normal observations.

1.4.1 Data-driven Demand Estimation

In practice we cannot expect to know beforehand the actual transformation that makes the demand stationary, since we only have access to a historical demand series $\{d_s\}_{s=1}^{t} = \{d_1, d_2, \ldots, d_t\}$ that is sampled from an unknown stochastic process (with unknown mean, variance, seasonality and functional form).

The strategy that we will implement consists of the following steps:

1. Propose a functional form rich enough to capture the demand within a well-defined family of functional forms.
2. Calibrate the parameters of the function that fit the demand closest via optimization.
3. Evaluate the performance of the proposed function out-of-sample.

For Step 1, we can use some of the modeling techniques that we have discussed in Section 1.3. For example, we can use for example the model presented in the previous section, $\hat{D}_{t+1} = \hat{A}h_t + \hat{\beta}D_t$. Here, the hats above the expression denote quantities that are calibrated from data. Recalling the definition of conditional expectation in Section 1.2.2, a reasonable approach to calculate the parameters for the demand is to solve the following optimization problem:

$$\min_{\hat{A},\hat{\beta}} \mathsf{E}[(\hat{D}_t(\hat{A}, \hat{\beta}) - D_t)^2], \tag{1.17}$$

This expression can be seen as an instance of the conditional expectation defined as the best predictor for a random variable. The solution of this problem, is in fact an approximation of the conditional expectation of the demand, that is, $\hat{D}_t(A^*, \beta^*) \approx \mathsf{E}[D_t | D_{t-1}, \ldots]$. The quality of the approximation depends crucially on two factors: That the functional form $\hat{D}_t$ allows to

approximate the conditional expectation. And also, on the sample size t of historical observations for the demand. Moreover, note that when the true parameters of the demand A and β are used, the difference $\hat{D}_t(A, \beta) - D_t \stackrel{\mathrm{d}}{=} V_t$. In general, $\mathsf{E}(V_t)$ should be equal to 0 (in terms of the base model, if V_t has non-zero mean c, define new variables $\alpha' = \alpha + c$ and $V_t' = V_t - c$, then $\mathsf{E}(V_t') = 0$). Therefore, this gives us another clue that $\mathsf{E}(\hat{D}_t(A^*, \beta^*) - D_t)$ should be approximately 0 when solving the optimization problem of fitting the demand.

To solve the optimization problem in Step 2, we solve the data-driven analogous of the problem in (1.17) with a sample of demand observations $\{d_s\}_{s=0}^t$ (in this case with the particular choice of demand function in Step 1):

$$\min_{\hat{A},\hat{\beta}} \frac{1}{t} \sum_{s=1}^{t} (\hat{A} h_{s-1} + \hat{\beta} d_{s-1} - d_s)^2, \tag{1.18}$$

This problem is a convex optimization program with a unique solution, that can be solved with any off-the-shelf quadratic optimization solver[8].

It is appealing to use a family of functions that can express very rich functional forms (for example, for a closed interval it is well-known that a Fourier basis can approximate any continuous function). The formalism would be to express the fitted demand $\hat{D}_t(\theta)$ as a function of parameters θ, for example, the weights of a deep neural network, or the collection of splits in a random forest. The approach would be the same, that is, to solve the optimization problem $\min_\theta \mathsf{E}[(\hat{D}_t(\theta) - D_t)^2]$.

The issue with this approach is that if the family of functions is too rich, the sample minimization would be too optimistic (potentially 0 error as $\hat{D}_t(\theta^*) = d_t$, for all observations in the sample $\{d_s\}_{s=1}^t$) and the performance for future predictions could be arbitrarily off. The two main reasons for this phenomenon are: First, in practice we only have access to *finite* samples that might not represent the full geometry of the conditional distribution (a well-known phenomenon in statistical analysis). Second, real-life stochastic process are in general changing in distribution over time (a so-called *doubly stochastic process*, where the original process changes distribution based on another unknown random variable). In the context of demand estimation, this could be a shock to the demand or a technological change. For example, the demand for horse carriages catastrophically decreased with the invention and mass production of mechanical engine cars (think that the measure of power of engines is called *horsepower*) in the early twentieth century. Likewise, the introduction of new competitors, or shocks to the demand or supply can change the trajectory of the demand.

While this problem cannot be fully mitigated, if there is enough data we can use what we have learned to come up with a partial solution: For the first problem of over-fitting the

[8] Linear least squares is an instance of Quadratic Programming. Note that in matrix form linear least squares has the objective $||\beta X - y||_2^2 = 2(\beta X X^T \beta/2 - \beta X y^T + y y^T/2)$ which is a quadratic program by letting $Q = XX^T/2$ and $c^T = -Xy^T$. Then, $\min_\beta ||\beta X - y||_2^2 = 2\min_\beta(\beta Q \beta^T + \beta c^T) + yy^T$.

demand, we can sub-divide our full sample of historical observation into multiple sub-samples where by setting up an experiment, we can use the LLN and the CLT to get an idea of the performance of our demand estimation strategy out-of-sample. For the second problem, imagine that distribution changes at random with a throw of a coin with small probability p, then, the time τ in between changes in distribution is in average $\mathsf{E}(\tau) = 1/p$. Then, if instead of training our model with all of the historical observations, we train the model with k observations, such that $k < \mathsf{E}(\tau)$, with probability $(1-p)^k \approx 1$ we are in the current distribution of the demand (another approach would be to weight the observations with a decaying weight). To see the point of using k observations imagine a sudden financial crisis that impacts the demand of luxury goods, such as designer bags. Using the whole historical demand will make the estimates to continue growing if the historical data includes most of the period before the crisis. Using the last k observation makes the estimates adjust faster to the new distribution of the demand.

Then, for Step 3 we take the following approach: Given historical observations of the demand $\{d_s\}_{s=1}^T$, a estimation window size k and a prediction horizon j. Define the sample optimization problem as $\ell(t,t') = 1/(t'-(t+1)) \sum_{s=t+1}^{t'} (Ah_{s-1} + \beta d_{s-1} - d_s)^2$. Likewise, define the absolute error $|\varepsilon_{t+j}(A,\beta)| = |\hat{D}_{t+j}(A,\beta) - d_{t+j}|$. The strategy is going to be to estimate multiple observations of the error out-of-sample, and average them as a measure of how the model performs out of sample. The pseudo-code routine to perform the *backtest* can be written as:

Algorithm 1 *Backtesting Demand Prediction*

```
Input: Historical observations for the demand d_1, ..., d_T. A training window k, and a
       prediction horizon j. And a model specification for the demand D̂_{t+j}(·)
Output: The average out-of-sample absolute error E(|ε_{t+j}|)
procedure BACKTEST({d_s}_{s=1}^T, k, j)
    μ_{|ε|} = 0
    for t = k to T − j do
        A*, β* ← argmin ℓ(t − k + 1, t)
        |ε_{t+j}(A*, β*)| = |D̂_{t+j}(A*, β*) − d_{t+j}|
        μ_{|ε|} = μ_{|ε|} + |ε_{t+j}(A*, β*)|/(T − k − j)
    end for
    return μ_{|ε|}
end procedure
```

With this procedure, given two models (say the linear one that we have been exploring) and a more sophisticated one like a Deep Neural Network model. The estimates of the mean absolute error μ_ε can be used to compare the performance of the model out-of-sample. This procedure is commonly used in finance to backtest algorithms. The key concept is that the measurement of the error is done outside of the training data of the model, thus, mimicking

the behavior of the model in real life. Likewise, the choosing of the training window size k can be evaluated with running this procedure multiple times with different sizes of k. Of course, to make the comparison fair, some exclusion of data might be needed as a larger window size needs more observations, and the number of observations for the error ε_{t+j} will differ depending on the window size. Lastly, this procedure can be easily modified to estimate the variance of the error or other metrics of interest.

The errors are themselves objects of interest. As discussed previously $V_{t+j} \stackrel{\mathrm{d}}{=} \hat{D}_{t+j}(A,\beta) - D_{t+j} = \varepsilon_{t+j}(A,\beta)$. Then, the mean $\mu_{\varepsilon(t+j)}$ and the variance of the errors $\sigma^2_{\varepsilon(t+j)}$ can be used to approximate the error of predictions. For example, approximating $V_{t+j} = \mu_{\varepsilon(t+j)} + \sigma_{\varepsilon(t+j)} Z$ allows to construct confidence intervals for the demand. For example, the probability that the forecasted demand at time $t+j$ exceeds the quantity x is equal to:

$$\mathsf{P}(D_{t+j} > x) = \mathsf{P}(\hat{D}_{t+j}(A,\beta) + V_{t+j} > x) = \mathsf{P}\left(Z > \frac{x - \hat{D}_{t+j}(A,\beta) - \mu_{\varepsilon(t+j)}}{\sigma_{\varepsilon(t+j)}}\right). \tag{1.19}$$

Which is the well-known tail cdf of a standard normal distribution $\bar{\Phi}\left(\frac{x-\hat{D}_{t+j}(A,\beta)-\mu_{\varepsilon(t+j)}}{\sigma_{\varepsilon(t+j)}}\right)$. It is important to note that the prediction horizon j is crucial, because in general the longer the prediction horizon, the larger the errors. That being said, there is no free lunch in prediction since the estimation of the mean error and its variance is in itself a random variable. Moreover, there is an implicit assumption that the distribution of the stochastic process is in some sense stationary or fixed within a time interval. In the real-world there is no guarantee that this is the case.

1.5 Exercises